Functions of Two Variables

Limits, Continuity and Partial Derivatives

Engineering Mathematics–I (CSE-1001)
BTech 1st Year, IIIT Bhopal

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To the Student

This is the second note of **Module 2**, and it is where the subject genuinely changes character. Everything in Module 1 concerned a function of one variable — a point on a line, approached from the left or from the right. Here the input is a point in a *plane*, and a plane can be approached from infinitely many directions, along infinitely many curves. Almost every surprise in this note comes from that one fact.

Read the companion note *The Mean Value Theorems* first if you have not; and keep the Module 1 note on *Sequences* nearby, because the Sandwich Theorem does as much work here as it did there.

A warning before you begin. Three results in this note look like their one-variable counterparts and are not. Partial derivatives can exist at a point where the function is not even continuous. The two mixed second derivatives f_{xy} and f_{yx} need not be equal. And checking every straight line through a point tells you nothing at all about whether a limit exists. Each of these is marked with a *Common mistake* box. Read those boxes twice.

How to use this note:

1. Read the matching section after each lecture, working every example yourself before reading the solution.
2. Stop at every *Try it yourself* box — there are seven.
3. When a counterexample appears, do not skim it. In this topic the counterexamples *are* the theory: they are what force the definitions to take the shape they do.
4. Attempt the companion question bank once a section is finished.

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1 What Changes When There Are Two Variables

A function of one variable, $f : \mathbb{R} \to \mathbb{R}$, takes a number and returns a number. A function of two variables takes a *point of the plane*:

$$f : D \subseteq \mathbb{R}^2 \to \mathbb{R}, \qquad (x, y) \mapsto f(x, y).$$

Its graph is the surface $z = f(x, y)$ sitting above the domain D.

There is a second, flatter way of picturing the same function, and it is the one that will matter most later. Fix a number c and mark every point of the domain at which $f(x, y) = c$. The resulting curve is a **level curve** of f; a family of them, drawn for several values of c, is a contour map of the surface. Figure 1 shows both pictures side by side.

Two variables is the normal case, not the exotic one

Almost nothing in engineering depends on one quantity alone. The temperature of a plate depends on *where* you measure it — two coordinates. The cost of manufacturing depends on labour and materials. The loss function of a neural network depends on every weight in the network, which is to say on millions of variables at once; and the training algorithm, gradient descent, is nothing but the two-variable theory of Section 5 scaled up. One-variable calculus is the special case. This note is where the general subject begins.

Here is the single structural difference that drives everything:

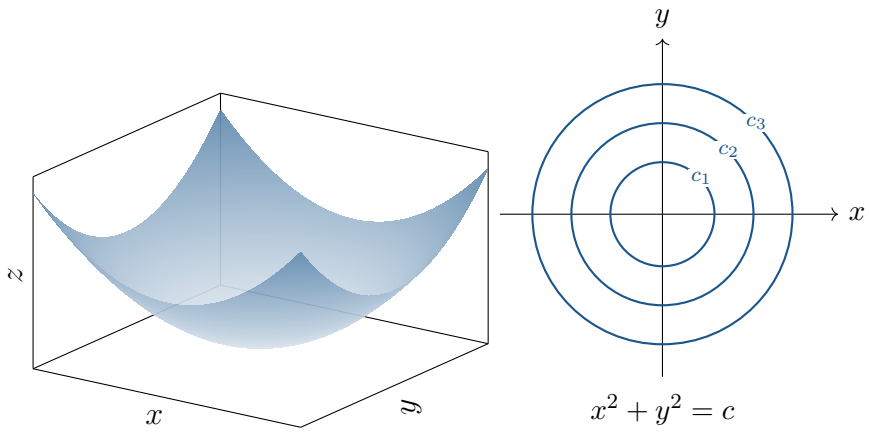


Figure 1: Two ways of seeing the same function $f(x, y) = x^2 + y^2$. On the left, the graph is a *surface* $z = f(x, y)$ in space. On the right, the domain is drawn flat and marked with *level curves* $f(x, y) = c$ — the contour lines of a map. The level curves are what you get by slicing the surface at constant height and looking down from above.

The approach problem

On the real line there are exactly **two** ways to approach a point a: from the left and from the right. That is why a one-variable limit exists precisely when two one-sided limits agree — a condition you can actually check.

In the plane there are **infinitely many** ways to approach a point (a, b): along any straight line through it, along any parabola, along any spiral. For the limit to exist, *every single one* of these must give the same value. You cannot check them one at a time.

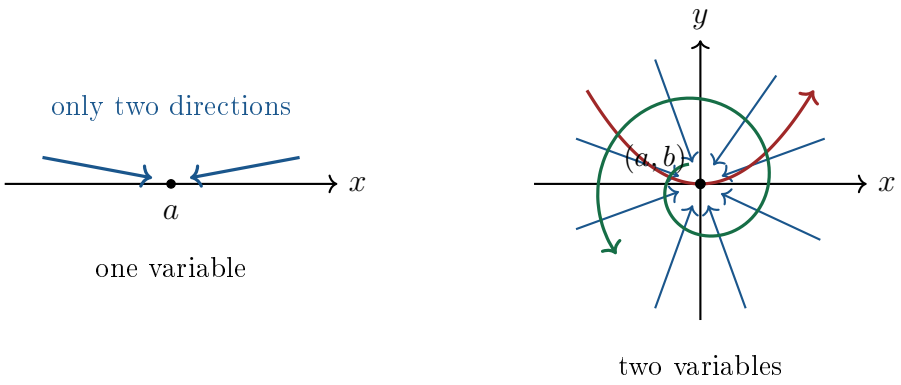


Figure 2: On the line, two approaches. In the plane, infinitely many — lines (blue), curves (red), spirals (green). The limit must agree along all of them.

2 Limits of Functions of Two Variables

Definition 2.1 (Limit)**.** Let f be defined on a disc around (a, b), except possibly at (a, b) itself. We say

$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \qquad \Longrightarrow \qquad |f(x,y) - L| < \varepsilon.$$

In words: the values of f can be forced within ε of L by confining (x, y) to a small enough punctured disc about (a, b) — a *disc*, not an interval, and that is the whole difference.

2.1 Proving a limit does not exist: the path principle

Proposition 2.2 (Path principle)**.** *If $f(x,y) \to L_1$ as $(x,y) \to (a,b)$ along one path, and $f(x,y) \to L_2$ along another path, with $L_1 \neq L_2$, then* $\lim_{(x,y)\to(a,b)} f(x,y)$ ***does not exist.***

This follows immediately from the definition: a single L would have to serve both paths.

Example 2.3. Show that $\lim_{(x,y)\to(0,0)} \dfrac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Solution. Approach along the x-axis, $y = 0$:

$$f(x,0) = \frac{x^2 - 0}{x^2 + 0} = 1 \longrightarrow 1.$$

Approach along the y-axis, $x = 0$:

$$f(0,y) = \frac{0 - y^2}{0 + y^2} = -1 \longrightarrow -1.$$

Two paths, two different values. By Proposition 2.2 the limit does not exist. □

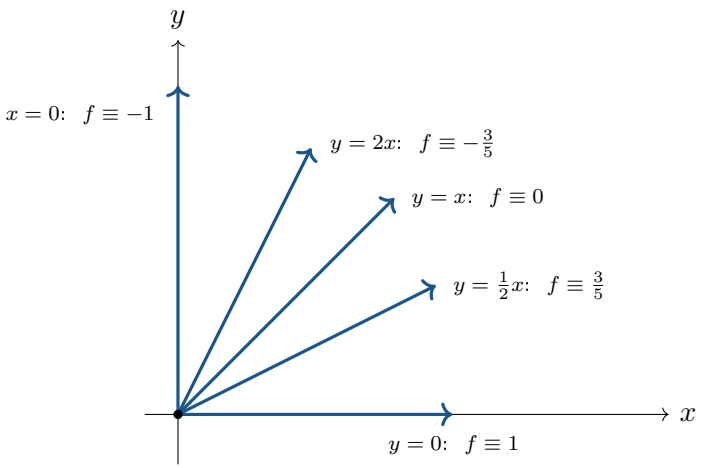


Figure 3: The path principle in action. Along the line $y = mx$ the function $f(x,y) = (x^2 - y^2)/(x^2 + y^2)$ is *constant*, equal to $(1 - m^2)/(1 + m^2)$, right up to the origin — and the vertical line $x = 0$ gives -1. Different lines give different constants, so no single number can be the limit. Notice also that *every* value between -1 and 1 occurs on some line: the function does not merely disagree on two paths, it disagrees on a continuum of them.

Figure 3 shows why this example is not a fluke. The function is constant along each ray from the origin, and the constant changes as the ray turns, so the "value at the origin" depends entirely on the direction of approach — which is precisely what a limit is not allowed to do.

2.2 The trap: paths can never prove a limit exists

Checking lines proves nothing

It is tempting to argue: "along $y = 0$ the limit is 0; along $x = 0$ it is 0; along every line $y = mx$ it is 0; therefore the limit is 0."
This reasoning is invalid. The straight lines are only some of the paths. A function can agree along every line through the origin and still have no limit, because some *curve* disagrees. The next example is exactly such a function, and it is the reason this box exists.
The path principle is a **one-way tool**: it can prove a limit *does not* exist, never that it does.

Example 2.4. Investigate $\lim_{(x,y)\to(0,0)} \dfrac{x^2y}{x^4+y^2}$.

Solution. **Along every straight line** $y = mx$ (with $m \neq 0$):

$$f(x, mx) = \frac{x^2 \cdot mx}{x^4 + m^2x^2} = \frac{mx^3}{x^2\left(x^2 + m^2\right)} = \frac{mx}{x^2 + m^2} \longrightarrow \frac{0}{m^2} = 0.$$

Along the axes the function is identically 0. So *every line through the origin gives the limit* 0.

Now approach along the parabola $y = x^2$**:**

$$f\left(x, x^2\right) = \frac{x^2 \cdot x^2}{x^4 + x^4} = \frac{x^4}{2x^4} = \frac{1}{2} \quad \text{for every } x \neq 0,$$

so along this path the function is *constantly* $\frac{1}{2}$ and tends to $\frac{1}{2}$.

Two paths give 0 and $\frac{1}{2}$. The limit **does not exist** — even though every straight line said it did. □

Why the parabola, and not a line?

The denominator contains x^4 and y^2. These two terms are comparable in size exactly when y is comparable to x^2 — so that is the curve along which the numerator and denominator can balance. When you hunt for a bad path, look at the *exponents* and choose $y = kx^p$ with p matching the denominator's own scaling.

2.3 Proving a limit does exist: squeeze and polar coordinates

Since paths cannot establish existence, we need a method that controls *all* approaches at once. Converting to polar coordinates does exactly that, because a single variable r then measures distance from the point regardless of direction.

Put $x = r\cos\theta$, $y = r\sin\theta$. Then $(x, y) \to (0, 0)$ is precisely $r \to 0^+$, and θ records the direction.

Example 2.5. Show that $\lim_{(x,y)\to(0,0)} \dfrac{x^2y}{x^2+y^2} = 0$.

Solution. In polar coordinates,

$$f = \frac{(r\cos\theta)^2(r\sin\theta)}{r^2} = r\cos^2\theta\,\sin\theta.$$

Therefore

$$|f| = r\left|\cos^2\theta\,\sin\theta\right| \le r,$$

because $|\cos^2\theta\sin\theta| \le 1$. The bound r depends on r **alone** and not on θ, so it controls every direction simultaneously. As $r \to 0$ we get $|f| \to 0$, hence by the Sandwich Theorem

$$\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^2+y^2} = 0. \qquad \square$$

The polar method has its own trap

The argument above works because the bound was r, *free of* θ. If you reach a bound such as r/θ or $\tan\theta$, which blows up for some directions, you have proved nothing: the convergence is not uniform in direction, and the limit may well fail to exist. Always check that the final bound involves r only.

Try it yourself

Evaluate, or show the limit does not exist:

$$\text{(i)}\ \lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}} \qquad \text{(ii)}\ \lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$$

Answers: (i) 0 *— in polars it is* $r\cos\theta\sin\theta$*, bounded by* r*. (ii) does not exist — it is* $\cos\theta\sin\theta$*, with no* r *at all, so the value depends only on direction:* 0 *along* $y = 0$ *and* $\frac{1}{2}$ *along* $y = x$*.*

3 Repeated (Iterated) Limits

There is a second, quite different thing one can compute: take the limits *one variable at a time.*

Definition 3.1 (Repeated limits)**.** The two **repeated** (or **iterated**) limits at (a, b) are

$$\lim_{x\to a}\left(\lim_{y\to b} f(x,y)\right) \qquad \text{and} \qquad \lim_{y\to b}\left(\lim_{x\to a} f(x,y)\right).$$

These are not the same object as the **double limit** $\lim_{(x,y)\to(a,b)} f(x,y)$ of Section 2, in which both variables move together.

Example 3.2. For $f(x,y) = \dfrac{x-y}{x+y}$, compute both repeated limits at the origin.

Solution. Inner limit in y first:

$$\lim_{y\to 0}\frac{x-y}{x+y} = \frac{x}{x} = 1 \quad (x \neq 0), \qquad \text{so} \qquad \lim_{x\to 0}\left(\lim_{y\to 0} f\right) = 1.$$

Inner limit in x first:

$$\lim_{x\to 0}\frac{x-y}{x+y} = \frac{-y}{y} = -1 \quad (y \neq 0), \qquad \text{so} \qquad \lim_{y\to 0}\left(\lim_{x\to 0} f\right) = -1.$$

The two repeated limits are 1 and -1. $\square$

Proposition 3.3. *If the two repeated limits exist and are* ***different****, the double limit cannot exist.*

So in the example above, $\lim_{(x,y)\to(0,0)} \frac{x-y}{x+y}$ does not exist. This is a genuinely useful test, and often the quickest one available.

3.1 The relationship between the three limits — stated correctly

A false statement found in many texts

The following claim appears in a great many sets of notes:

> *"If the double limit exists, then both repeated limits exist and are equal to it."*

This is false. The existence of the double limit says nothing about whether the *inner* limit of a repeated limit exists at all.

Example 3.4 (The counterexample)**.** Let

$$f(x,y) = \begin{cases} x \sin \dfrac{1}{y}, & y \neq 0, \\ 0, & y = 0. \end{cases}$$

Solution. **The double limit exists and is** 0**.** For all (x,y),

$$|f(x,y)| \leq |x| \leq \sqrt{x^2+y^2},$$

so $f \to 0$ as $(x,y) \to (0,0)$ by the Sandwich Theorem.

But one repeated limit does not exist. Fix any $x \neq 0$ and let $y \to 0$. Then $\sin(1/y)$ oscillates through every value in $[-1,1]$ without settling — taking the values $0, 1, 0, -1, \ldots$ along $y = \frac{1}{2\pi}, \frac{1}{2.5\pi}, \frac{1}{3\pi}, \frac{1}{3.5\pi}, \ldots$ — so $\lim_{y\to 0} f(x,y)$ **does not exist**. The repeated limit $\lim_{x\to 0}\big(\lim_{y\to 0} f\big)$ therefore does not exist either.

The double limit exists; a repeated limit does not. The claim in the box above is contradicted. □

Here is the correct statement.

Theorem 3.5 (Double and repeated limits)**.** *Suppose the double limit*

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

exists. If a repeated limit also exists, then it equals L. In particular, if all three exist, all three are equal.

The converse fails too, and for a different reason: the repeated limits can both exist and agree while the double limit does not.

Example 3.6. For $f(x,y) = \dfrac{x^2 y}{x^4+y^2}$ of Example 2.4, both repeated limits at the origin equal 0:

$$\lim_{y\to 0} f(x,y) = \frac{0}{x^4} = 0 \quad (x \neq 0), \qquad \lim_{x\to 0} f(x,y) = \frac{0}{y^2} = 0 \quad (y \neq 0),$$

so both repeated limits are 0. Yet we showed the double limit does not exist, because the parabola $y = x^2$ gives $\frac{1}{2}$.

What you observe	What you may conclude	Why
Repeated limits differ	Double limit does not exist	Prop. 3.3
Repeated limits agree	**Nothing**	$\frac{x^2y}{x^4+y^2}$
Two paths disagree	Double limit does not exist	Prop. 2.2
All tested paths agree	**Nothing**	$\frac{x^2y}{x^4+y^2}$
Polar bound $\leq g(r) \to 0$	Double limit exists, $= L$	Sandwich

The asymmetry to remember

Every cheap test in this topic can only ever give a **negative** answer. To prove a limit *exists* you must bound the function in a way that is uniform over all directions — which in practice means the polar/squeeze method or the ε–δ definition itself.

4 Continuity

Definition 4.1 (Continuity at a point)**.** f is **continuous at** (a, b) if
(i) $f(a, b)$ is defined;
(ii) $\lim_{(x,y)\to(a,b)} f(x, y)$ exists;
(iii) the two are equal: $\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b)$.
f is continuous on a region if it is continuous at every point of it.

The structure is identical to the one-variable definition; only the meaning of "$\to$" has changed.

Example 4.2. Is

$$f(x,y) = \begin{cases} \dfrac{x^2y}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0) \end{cases}$$

continuous at the origin?

Solution. $f(0, 0) = 0$ is defined; we showed in Section 2 that the limit is 0; and the two agree. So **yes**, f is continuous at the origin. □

Example 4.3. Is $g(x, y) = \dfrac{x^2 - y^2}{x^2 + y^2}$ (with $g(0, 0) = 0$) continuous at the origin?

Solution. **No.** The limit does not exist (Section 2), so condition (ii) fails. No choice of value at the origin can repair this — the discontinuity is not removable. □

	One variable	Two variables
Approaches to check	Two: left and right	Infinitely many directions and curves
Test for a limit	Compare two one-sided limits	No finite test; use squeeze or ε–δ
Disproving a limit	One-sided limits differ	Two paths differ
Typical difficulty	Routine	The hard part of the topic

5 Partial Derivatives

If a function depends on two variables, we can vary one and hold the other fixed. That is the whole idea.

Definition 5.1 (Partial derivatives)**.** For $z = f(x, y)$,

$$\frac{\partial f}{\partial x}(x, y) = \lim_{h \to 0} \frac{f(x + h,\, y) - f(x, y)}{h}, \qquad \frac{\partial f}{\partial y}(x, y) = \lim_{k \to 0} \frac{f(x,\, y + k) - f(x, y)}{k},$$

provided the limits exist.

Notation. All of these mean the same thing:

$$\frac{\partial f}{\partial x} = \frac{\partial z}{\partial x} = f_x(x, y) = D_x f = z_x.$$

What a partial derivative measures

Slice the surface $z = f(x, y)$ with the vertical plane $y =$ constant. The slice is an ordinary curve in one variable, and f_x is its ordinary slope. A partial derivative is a one-variable derivative taken along a slice — which is why every differentiation rule you already know still applies, with the other variable treated as a constant.

5.1 From first principles

Example 5.2. Find f_x and f_y from the definition for $f(x, y) = x^2 y + y^3$.

Solution.

$$f_x = \lim_{h \to 0} \frac{\left[(x + h)^2 y + y^3\right] - \left[x^2 y + y^3\right]}{h} = \lim_{h \to 0} \frac{y\,(2xh + h^2)}{h} = \lim_{h \to 0} y(2x + h) = 2xy.$$

$$f_y = \lim_{k \to 0} \frac{\left[x^2(y + k) + (y + k)^3\right] - \left[x^2 y + y^3\right]}{k} = \lim_{k \to 0} \frac{x^2 k + 3y^2 k + 3yk^2 + k^3}{k} = x^2 + 3y^2.$$

□

In practice one differentiates directly, treating the other variable as a constant:

$$\frac{\partial}{\partial x}\left(x^2 y + y^3\right) = 2xy,$$

since y^3 is constant as far as x is concerned.

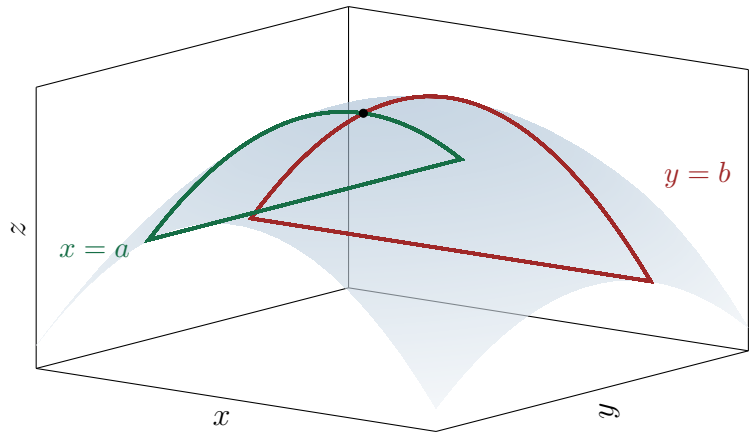


Figure 4: What a partial derivative measures. Freezing $y = b$ cuts the surface in the red curve, and the slope of that curve at the marked point is $f_x(a, b)$. Freezing $x = a$ cuts it in the green curve, whose slope there is $f_y(a, b)$. Each partial derivative is an ordinary one-variable derivative, taken along one slice — which is why all the familiar differentiation rules still apply.

5.2 Three or more variables

Nothing changes except the bookkeeping. For $w = f(x, y, z)$ we hold two variables fixed and vary the third.

Example 5.3. For $f(x, y, z) = x^2y + yz^3 + xz$, find all three first-order partials.

Solution.

$$f_x = 2xy + z, \qquad f_y = x^2 + z^3, \qquad f_z = 3yz^2 + x. \qquad \square$$

Example 5.4. For $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ show that $f_x = x/f$, and write down f_y and f_z.

Solution. By the chain rule, $f_x = \dfrac{2x}{2\sqrt{x^2 + y^2 + z^2}} = \dfrac{x}{f}$, and by symmetry $f_y = y/f$, $f_z = z/f$. $\square$

Try it yourself

Find all first-order partials of (i) $f = e^{xyz}$ and (ii) $f = \ln(x^2 + y^2 + z^2)$.

Answers: (i) $yz\,e^{xyz}$, $xz\,e^{xyz}$, $xy\,e^{xyz}$. *(ii)* $\dfrac{2x}{x^2 + y^2 + z^2}$, *and similarly in* y *and* z.

5.3 Higher-order partial derivatives

Differentiating again produces four second-order partials:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \qquad f_{xy} = (f_x)_y, \qquad f_{yx} = (f_y)_x.$$

Fix your convention and keep it

The subscript notation is read **left to right in the order the differentiations are performed**: f_{xy} means "differentiate by x first, then by y". The operator notation runs the opposite way:

$$f_{xy} = \frac{\partial^2 f}{\partial y\, \partial x}.$$

Different books make different choices. Pick one — we use the one above throughout — and never mix the two inside a single calculation. Most errors in the next subsection are bookkeeping errors of exactly this kind.

Example 5.5. For $f(x, y, z) = \ln(xy + z^2)$, find f_y and then f_{yx}.

Solution.

$$f_y = \frac{x}{xy + z^2}.$$

Now differentiate with respect to x — and note that x appears in *both* the numerator and the denominator, so the quotient rule is needed in full:

$$f_{yx} = \frac{\partial}{\partial x}\left(\frac{x}{xy + z^2}\right) = \frac{(xy + z^2)\cdot 1 - x \cdot y}{(xy + z^2)^2} = \frac{z^2}{(xy + z^2)^2}.$$

□

The slip this example is built around

A very common wrong answer here is

$$f_{yx} = \frac{(xy + z^2)\cdot 0 - x \cdot y}{(xy + z^2)^2} = \frac{-xy}{(xy + z^2)^2},$$

obtained by treating the numerator x as a constant while differentiating with respect to x. It is wrong in both sign and value. And note that it cannot be caught by the usual symmetry check — $-xy/(xy + z^2)^2$ is also symmetric in x and y, so it "passes" Clairaut's test while being wrong. There is no substitute for doing the differentiation carefully.

5.4 Must the mixed partials be equal?

For every function you are likely to meet, $f_{xy} = f_{yx}$. But this is a theorem with hypotheses, not an identity — and it can fail.

Example 5.6 (Mixed partials that disagree)**.** Let

$$f(x, y) = \begin{cases} \dfrac{xy\,(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Show that $f_{xy}(0, 0) = -1$ while $f_{yx}(0, 0) = +1$.

Solution. Work from the definition at the origin, one step at a time.

Step 1 — f_x on the y-axis. For $y \neq 0$,

$$f_x(0,y) = \lim_{h\to 0} \frac{f(h,y) - f(0,y)}{h} = \lim_{h\to 0} \frac{1}{h} \cdot \frac{hy\,(h^2 - y^2)}{h^2 + y^2} = \frac{y\,(-y^2)}{y^2} = -y,$$

and $f_x(0,0) = 0$ as well, so $f_x(0,y) = -y$ for all y.

Step 2 — f_y on the x-axis. Symmetrically, for $x \neq 0$,

$$f_y(x,0) = \lim_{k\to 0} \frac{1}{k} \cdot \frac{xk\,(x^2 - k^2)}{x^2 + k^2} = \frac{x\,(x^2)}{x^2} = x,$$

so $f_y(x,0) = x$ for all x.

Step 3 — differentiate once more.

$$f_{xy}(0,0) = \left.\frac{\partial}{\partial y}\right|_{y=0} f_x(0,y) = \left.\frac{\partial}{\partial y}\right|_{y=0} (-y) = -1,$$

$$f_{yx}(0,0) = \left.\frac{\partial}{\partial x}\right|_{x=0} f_y(x,0) = \left.\frac{\partial}{\partial x}\right|_{x=0} (x) = +1.$$

The two mixed partials differ. □

Which one is -1?

Under the convention $f_{xy} = (f_x)_y$ fixed above, the answer is $f_{xy}(0,0) = -1$ and $f_{yx}(0,0) = +1$. Under the opposite convention the labels swap. The numbers ± 1 are not in doubt; which label carries which sign depends entirely on the convention, so **state your convention before you start**. A solution that writes $f_{xy}(0,0) = +1$ without saying which order it means has not answered the question.

5.5 Clairaut's Theorem

What rules this behaviour out is continuity of the mixed partials.

Theorem 5.7 (Clairaut / Schwarz)**.** *If f is defined on a disc containing (a,b) and both f_{xy} and f_{yx} are **continuous** on that disc, then*

$$f_{xy}(a,b) = f_{yx}(a,b).$$

Idea of the proof. Form the second difference

$$\Delta = f(a+h, b+k) - f(a+h, b) - f(a, b+k) + f(a,b).$$

Grouping the four terms one way and applying the one-variable Mean Value Theorem twice gives $\Delta = f_{xy}(p,q)\,hk$ for some point (p,q) in the rectangle; grouping them the other way gives $\Delta = f_{yx}(p',q')\,hk$ for another point. Dividing by hk and letting $(h,k) \to (0,0)$, both interior points approach (a,b), and continuity of the two mixed partials forces the two limits — hence $f_{xy}(a,b)$ and $f_{yx}(a,b)$ — to coincide. □

Why Example 5.6 is not a contradiction

In that example f_{xy} and f_{yx} both exist everywhere, but they are **not continuous at the origin** — indeed they take different values there. The hypothesis of Theorem 5.7 fails, so the theorem has nothing to say, and no contradiction arises. This is exactly how a counterexample and a theorem are supposed to fit together.

6 A Warning: Partials Can Exist Without Continuity

In one variable, differentiability at a point forces continuity there. In two variables the analogous statement with *partial* derivatives is false, and spectacularly so.

Example 6.1. Let

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Show that $f_x(0,0)$ and $f_y(0,0)$ both exist, yet f is not continuous at the origin.

Solution. **The partials exist.** Along the axes the function is identically zero: $f(h,0) = 0$ and $f(0,k) = 0$. Hence

$$f_x(0,0) = \lim_{h\to 0} \frac{0-0}{h} = 0, \qquad f_y(0,0) = \lim_{k\to 0} \frac{0-0}{k} = 0.$$

But f is not continuous there. Along $y = 0$, $f = 0$; along $y = x$,

$$f(x,x) = \frac{x^2}{2x^2} = \frac{1}{2} \quad (x \neq 0).$$

Two paths give 0 and $\frac{1}{2}$, so the limit does not exist and f is discontinuous at the origin. □

Why this had to happen

$f_x(0,0)$ and $f_y(0,0)$ are computed from the behaviour of f along *two lines only* — the x-axis and the y-axis. The function is free to do anything it likes off those two lines, and this one does. Partial derivatives see two directions; continuity is about all of them.

The consequence for the whole module

"Both partial derivatives exist" is therefore a **much weaker** condition than one-variable differentiability. It is too weak to be the definition of differentiability in two variables, and the next note (*Differentiability, Chain Rule and Gradient*) builds the correct definition — one that *does* imply continuity. Everything that goes wrong in this section is the motivation for that definition.

7 Homogeneous Functions and Euler's Theorem

One class of functions has partial derivatives tied together by a single identity, and it appears often enough to be worth a short section.

Definition 7.1 (Homogeneous function)**.** $f(x,y)$ is **homogeneous of degree** n if

$$f(tx,\, ty) = t^n f(x,y) \qquad \text{for all } t > 0.$$

Scaling both inputs by t scales the output by t^n. For example $f = x^3 + 3x^2y + y^3$ is homogeneous of degree 3; $f = \frac{x^3+y^3}{x+y}$ of degree 2; and $f = \tan^{-1}(y/x)$ of degree 0, since the ratio y/x is unchanged by scaling.

Theorem 7.2 (Euler's Theorem)**.** *If f is homogeneous of degree n and has continuous first partials, then*

$$x\,\frac{\partial f}{\partial x} + y\,\frac{\partial f}{\partial y} = n\,f.$$

Proof. Differentiate the identity $f(tx,ty) = t^n f(x,y)$ with respect to t, holding x and y fixed. By the chain rule the left side gives

$$x\,f_x(tx,ty) + y\,f_y(tx,ty),$$

and the right side gives $nt^{n-1}f(x,y)$. Now set $t = 1$:

$$xf_x(x,y) + yf_y(x,y) = nf(x,y). \qquad \square$$

Example 7.3. Verify Euler's theorem for $f = x^3 + 3x^2y + y^3$.

Solution. Here $n = 3$. We have $f_x = 3x^2 + 6xy$ and $f_y = 3x^2 + 3y^2$, so

$$xf_x + yf_y = 3x^3 + 6x^2y + 3x^2y + 3y^3 = 3\left(x^3 + 3x^2y + y^3\right) = 3f. \qquad \square$$

Corollary 7.4 (Second-order form)**.** *Under the same hypotheses, with continuous second partials,*

$$x^2 f_{xx} + 2xy\,f_{xy} + y^2 f_{yy} = n(n-1)\,f.$$

Proof. Differentiate $xf_x + yf_y = nf$ partially with respect to x and multiply by x; do the same with y; add, and use $f_{xy} = f_{yx}$. $\square$

Try it yourself

Verify Euler's theorem for $f = \tan^{-1}\left(\dfrac{y}{x}\right)$.

Answer: f is homogeneous of degree 0, so the theorem predicts $xf_x + yf_y = 0$. Indeed $f_x = \frac{-y}{x^2+y^2}$ and $f_y = \frac{x}{x^2+y^2}$, and $xf_x + yf_y = \frac{-xy+xy}{x^2+y^2} = 0$.

8 Summary

The five things to carry forward

1. A limit in two variables must agree along **every** approach. Disagreement along two paths disproves existence; agreement along any number of paths proves nothing.
2. To prove a limit exists, bound $|f - L|$ by a quantity depending on r alone (polar coordinates), then squeeze.

3. Repeated limits are a different object from the double limit. Unequal repeated limits disprove the double limit; equal ones prove nothing; and the double limit can exist when a repeated limit does not.

4. A partial derivative is an ordinary derivative along a slice. Mixed partials agree when they are continuous (Clairaut), and can genuinely disagree when they are not.

5. **Existence of both partial derivatives does not imply continuity.** This is the central surprise of the note, and the reason the next note needs a stronger notion of differentiability.

Looking ahead. The next note asks what "differentiable" ought to mean for $f(x, y)$. The answer is not "both partials exist" — we have just seen that fail — but a condition saying the surface is well approximated by a *plane*. From it follow the chain rule, the gradient, tangent planes and linearisation.

Companion material: The Mean Value Theorems (Module 2, Note 1); Functions of Two Variables — Question Bank.